

Category	Method	What it is
Data Augmentation	EDA	Uses dictionary-based synonym replacements, random insertion, random swap, and random deletion
	TextCut	Randomly replaces small parts of the BERT representation of one instance with those of another
	ECRT	Learns to map encoder representations to a new space whose components are independent given the class, enabling generation of new minority examples by permuting/sampling components
Loss Functions	FL (Focal Loss)	Down-weights instances the model is already confident on
	ADL	Extends Dice loss (which captures class-wise F1) with confidence-based down-weighting of easy predictions
	LDAM	Uses larger class-dependent margins for minority classes to improve their separation and generalization
	WBCE	Assigns higher weights to minority classes, but also increases the weight of their negative instances
	DB (Distribution-Balanced)	Adds “Negative Tolerant Regularization” — caps the loss for negative classes once the model is already confident, instead of letting the punishment build up indefinitely
Staged Learning	cRT	Two-stage re-training: normal training first, then class-balanced sampling in a second stage after freezing the representation weights
	ST (Sequential Targeting)	Models imbalanced classification as a continual-learning task with stages where the data gradually becomes more balanced
Model Design	tau-norm	Normalizes classifier weights using a temperature hyperparameter tau
	SetConv	Learns class representatives from support sets using convolution kernels to capture intra/inter-class correlations
	ProtoBERT	Uses class centroids in a learned BERT feature space, classifies by similarity to class representatives
	HSCNN	Uses class representatives only for tail classes; head classes go through a standard text CNN